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Roof Separation Index

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Separation Index (RSI), PVK Panel 21, SCCL

	Thickness (m)	Rock Type	RQD (%)	Sc (MPa)	St (MPa)
Overburden	166.34	Overburden	76	12.5	1.1
Main Roof	7.66	Grey and Carb. Clay	50	12.5	1.1
	8.28	Cgsst, GW	80	12.5	1.1
	2.00	Grey and Carb. Clay	50	12.5	1.1
	12.49	Cg-sst, GW, Pebble	78	8.7	0.9
Immediate Roof	6.53	Cg-fgsst	94	12.9	1.2
	3.70	Cg-sst, GW, Pebble	77	8.7	0.9
	4.00	Cg-sst, GW, Pebble	93	8.7	0.9
	1.30	Cgsst, GW	44	12.5	1.1
Extraction height	3.00	Coal	40	23.1	1.5

Rock type	Thickness (m)	RQD, %	σ_c , Mpa	σ_t , Mpa	Density, Kg/m ³	Emod, GPa	ACL, cm
Coal	3	40	23.16	1.54	1400	2	7.70
Cgsst, GW	1.3	44	12.46	1.07	2050	3.86	8.67
CGsst, GW, Pebble	4	93	8.68	0.9	2040	2.69	24.27
Cgsst, GW, Pebble	3.7	77	8.68	0.9	2040	2.69	13.70
Cg to Fg sst	6.53	94	12.93	1.15	2060	4.01	19.2
Cgsst, GW, Pebble	12.49	78	8.68	0.9	2040	2.69	16.6
Grey and carbonaceous clay	2	50	12.46	1.07	2050	3.86	7.7
Cgsst, GW	8.28	80.3	12.46	1.07	2050	3.86	23.65
Grey and carbonaceous clay	7.66	50	12.46	1.07	2050	3.86	7.7
Cgsst, GW	8.54	69	12.46	1.07	2050	3.86	14.1
Cgsst, GW	7.3	85	12.46	1.07	2050	3.86	32.14
Overburden	150.5						

Step I: From the table of rating, find RSI for each layer.

Roof Separation Index (RSI)

Sl. No.	Parameter	Unit/ Rating	Rating distribution				
1	Uni-axial compressive strength	MPa	<5	5-25	25-50	>50	
		Rating	1	2	4	7	
2	RQD	%	<25	25-50	50-75	75-90	>90
		Rating	3	8	13	17	20
3	Average core length	cm	<5	5-30	30-100	100-300	
		Rating	5	10	20	25	
4	Ground water condition/wetness of strata	Condition	Dry	Watery formation			
		Rating	10	0			
5	Bed thickness	m	<1.5	1.5-2.5	2.5-3.5	3.5-4.5	>=4.5
		Rating	-28	-16	-12	-5	0
6	Rock formation of bed	Rock type	Sandstone	Others (shale, coal, clay, shaly coal)			
		Rating	0	-25			

Table of Rating

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Roof Separation Index (RSI)

Rock type	Thickness (m)	RQD, %	σ_c , Mpa	σ_t , Mpa	Density, Kg/m ³	Emod, GPa	ACL, cm	σ_c factor	RQD factor	ACL factor	Factor for presence of ground water	adjustment factor for thickness	adjustment factor for type of rock formation	Roof separation Index, RSI
Coal	3	40	23.16	1.54	1400	2	7.70	2	8	10	10	-12	-25	-7
Cgsst, GW	1.3	44	12.46	1.07	2050	3.86	8.67	2	8	10	10	-28	0	2
CGsst, GW, Pebble	4	93	8.68	0.9	2040	2.69	24.27	2	20	10	10	-5	0	37
Cgsst, GW, Pebble	3.7	77	8.68	0.9	2040	2.69	13.70	2	17	10	10	-5	0	34
Cg to Fg sst	6.53	94	12.93	1.15	2060	4.01	19.2	2	20	10	10	0	0	42
Cgsst, GW, Pebble	12.49	78	8.68	0.9	2040	2.69	16.6	2	17	10	10	0	0	39
Grey and carbonaceous clay	2	50	12.46	1.07	2050	3.86	7.7	2	13	10	10	-16	-25	-6
Cgsst, GW	8.28	80.3	12.46	1.07	2050	3.86	23.65	2	17	10	10	0	0	39
Grey and carbonaceous clay	7.66	50	12.46	1.07	2050	3.86	7.7	2	13	10	10	0	-25	10
Cgsst, GW	8.54	69	12.46	1.07	2050	3.86	14.1	2	13	10	10	0	0	35
Cgsst, GW	7.3	85	12.46	1.07	2050	3.86	32.14	2	17	20	10	0	0	49
Overburden	150.5													

For Example :

Coal layer

	Parameter	Given Value	Range	Rating
①	UCS	23.16	5-25	2
②	RQD	40 %	25-50	8
③	Avg. core length	7.70 cm	5~30	10
④	Ground water	Take always dry unless said		10
⑤	Bed thickness	3m	2.5~3.5m	-12
⑥	Rock formation	Coal	Coal	-25

$$RSI = 2 + 8 + 10 + 10 - 12 - 25$$

$$= -7$$

Matching

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Do this for all the layers of the table.
This was step-I.

step-II : Make a table of Four rows

(i) Coal

(ii) Immediate Roof (IR)

Immediate Roof are the layers above the coal layer & first occurrence of a clay layer.

(ii) Main Roof (MR)

Main Roof are the layers above Immediate Roof and then the occurrence of another clay layer.

(iv) Overburden (Remaining above MR)

Roof Separation Index (RSI)

	Rock type	Thickness (m)	RQD, %	σ_c , Mpa	σ_c , Mpa	Density, Kg/m ³	Emod, GPa	ACL, cm	σ_c factor	RQD factor	ACL factor	Factor for presence of ground water	adjustment factor for thickness	adjustment factor for type of rock formation	Roof separation Index, RSI	
IR	Coal	3	40	23.16	1.54	1400	2	7.70	2	8	10	10	-12	-25	-7	0
	Cgsst, GW	1.3	44	12.46	1.07	2050	3.86	8.67	2	8	10	10	-28	0	2	1
	CGsst, GW, Pebble	4	93	8.68	0.9	2040	2.69	24.27	2	20	10	10	-5	0	37	2
	Cgsst, GW, Pebble	3.7	77	8.68	0.9	2040	2.69	13.70	2	17	10	10	-5	0	34	3
	Cg to Fg sst	6.53	94	12.93	1.15	2060	4.01	19.2	2	20	10	10	0	0	42	4
	Cgsst, GW, Pebble	12.49	78	8.68	0.9	2040	2.69	16.6	2	17	10	10	0	0	39	5
MR	Grey and carbonaceous clay	2	50	12.46	1.07	2050	3.86	7.7	2	13	10	10	-16	-25	-6	6
	Cgsst, GW	8.28	80.3	12.46	1.07	2050	3.86	23.65	2	17	10	10	0	0	39	7
	Grey and carbonaceous clay	7.66	50	12.46	1.07	2050	3.86	7.7	2	13	10	10	0	-25	10	8
OB	Cgsst, GW	8.54	69	12.46	1.07	2050	3.86	14.1	2	13	10	10	0	0	35	9
	Cgsst, GW	7.3	85	12.46	1.07	2050	3.86	32.14	2	17	20	10	0	0	49	10
	Overburden	150.5														11

Layers

Coal	0
IR	1, 2, 3, 4, 5, 6
MR	7, 8
OB	9, 10, 11

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Layer	Thickness, m	Density, Kg/m ³	RQD, %	Comp strength, MPa	Tensile strength, MPa
Coal	3	1400	40	23.16	1.54
Immediate roof layer	30.02	2045	80	10.02	0.97
Main roof layer	15.94	2050	66	12.46	1.07
Overburden	166.34	2050	76	12.46	1.07

Thickness = Add all the layers that fall under corresponding group

For example ; IR (Immediate Roof) :-

$$\begin{aligned}\text{Thickness} &= \text{Layer 1} + 2 + 3 + 4 + 5 + 6 \\ &= 1.3 + 4 + 3.7 + 6.53 + 12.49 + 2 \\ &= 30.02 \text{ m}\end{aligned}$$



For density, RQD, σ_c (intact), σ_t (intact)
Take weighted average

For example ; RQD for Immediate Roof ;

$$\text{RQD for IR} = \frac{\sum (\text{thickness of layer})_i \times \text{RQD}_i}{\sum (\text{thickness of layer})_i}$$

$$\begin{aligned}\text{RQD for IR} &= \frac{(1.3 \times 44) + (4 \times 93) + (3.7 \times 77) + (6.53 \times 94) + (12.49 \times 78) + (2 \times 50)}{(1.3 + 4 + 3.7 + 6.53 + 12.49 + 2)} \\ &= 80\end{aligned}$$

DO same for density, σ_c (intact), σ_t (intact)

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Layer	Thickness, m	Density, Kg/m ³	RQD, %	Comp strength, MPa	Tensile strength, MPa
Coal	3	1400	40	23.16	1.54
Immediate roof layer	30.02	2045	80	10.02	0.97
Main roof layer	15.94	2050	66	12.46	1.07
Overburden	166.34	2050	76	12.46	1.07

σ_c (rock mass) σ_t (rock mass)

In this table, also add two more columns σ_c (rock mass) & σ_t (rock mass)

For calculating value of σ_c (rock mass) & σ_t (rock mass) use this formula :

$$\sigma_c (\text{rock mass}) = \sigma_c (\text{intact}) \times 0.5 \times \frac{\text{RQD}}{100}$$

For example ;

$$\sigma_c (\text{rock mass}) \text{ of coal} = 23.16 \times 0.5 \times \frac{40}{100}$$

$$= 4.632 \text{ MPa}$$

$$\sigma_c (\text{rock mass}) \text{ of IR} = 10.02 \times 0.5 \times \frac{80}{100}$$

$$= 4.008 \text{ MPa}$$

Hence we get new table as ;

Layer	Thickness, m	Density, Kg/m ³	RQD, %	Comp strength, MPa	Tensile strength, MPa	σ_c (rock mass)	σ_t (rock mass)
Coal	3	1400	40	23.16	1.54	4.632	0.308
Immediate roof layer	30.02	2045	80	10.02	0.97	4.008	0.388
Main roof layer	15.94	2050	66	12.46	1.07	4.118	0.352
Overburden	166.34	2050	76	12.46	1.07	4.739	0.406

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Step - II : Calculate span of Local Fall, Subsequent Local fall, Main Fall, Periodic Caving

Beam Model Concept



Support from both ends

Cantilever Model Concept



Support from one end

Longwall mining :

① Local fall : For the first time the fall will occur (coal height) extraction.



Immediate roof support on back and front as well.

(Beam Model)

② Subsequent Local fall : When local fall occur, the roof supports on PRS and on the face of the longwall.
(Cantilever Model)

Similarly, this is the concept for Main roof.

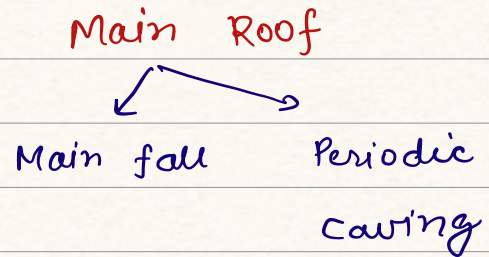
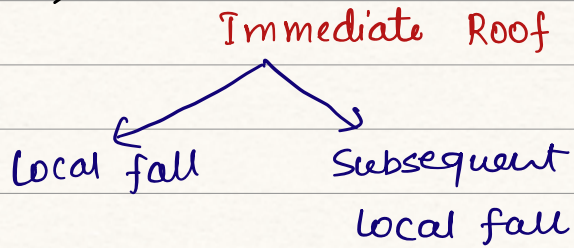
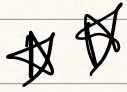
Main fall $\longrightarrow$ Beam Model

Periodic fall $\longrightarrow$ Cantilever Model

$$\text{Span in Beam Model (m)} = \sqrt{\frac{2 \times \sigma_t (\text{rockman}) \times \text{thickness}}{\gamma (\text{unit weight})}}$$

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$$\text{Span in cantilever model (m)} = \sqrt{\frac{\sigma_t (\text{rockmass}) \times \text{thickness}}{\gamma (\text{unit weight})}}$$



Calculate span :-

$$\text{span (Local Fall)} = \sqrt{\frac{2 \times 0.388 \times 30.02 \text{ IR}}{8 \times 9 \times 10^{-6} \text{ (MPa/m)}}$$

$$= \sqrt{\frac{2 \times 0.388 \times 30.02}{2045 \times 10 \times 10^{-6} \text{ IR}}}$$

$$= 33.75 \text{ m}$$

$$\text{span (SLF)} = \sqrt{\frac{0.388 \times 30.02 \text{ IR}}{2045 \times 10 \times 10^{-6} \text{ IR}}}$$

$$= 13.77 \text{ m}$$

★ Add the span (cumulative)

$$\text{Span} = 33.75 + 13.77 = 47.52 \text{ m}$$

Main Roof span

$$\begin{aligned} \text{Span (Main fall)} &= \sqrt{\frac{2 \times 0.352 \times 15.94 \text{ MR}}{2050 \times 10 \times 10^{-6}}} \\ &= 23.4 \text{ m} \end{aligned}$$

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$$\begin{aligned}\text{Cumulative span} &= 47.52 + 23.4 \\ &= 70.92 \text{ m}\end{aligned}$$

$$\begin{aligned}\text{Span (Periodic Caving)} &= \sqrt{\frac{0.352 \times 15.94}{2050 \times 10 \times 10^{-6}}} \text{ MR} \\ &= 9.55 \text{ m}\end{aligned}$$

$$\begin{aligned}\text{Cumulative span} &= 70.92 + 9.55 \\ &= 80.47 \text{ m}\end{aligned}$$

step - IV : Calculate W_s

$$W_s = 324 \times (FL)^{-0.48} \times (D)^{-1.38} \times \sqrt{\frac{\sigma_{mIR} \times T_{IR}}{\gamma_{IR}}} \times \sqrt{\frac{\sigma_{mMR} \times T_{MR}}{\gamma_{MR}}}$$

$W_s \rightarrow$ weighing span / main fall span (m)

$FL \rightarrow$ face length (m)

$D \rightarrow$ depth of working (m)

$$\sigma_{mIR} = \left(\frac{\sigma_{Hmean} + \sigma_t (\text{rock mass})}{100} \right) \times RQD$$

$$\sigma_{Hmean} = \left(\frac{\sigma_{Htop} + \sigma_{Hbottom}}{2} \right)$$

$$\sigma_H = \frac{\nu}{1-\nu} \times \sigma_v + \frac{BEG \times (1000+h)}{1-\nu}$$

$\nu \rightarrow$ poisson's ratio (0.25)

$$\sigma_v = \gamma D \quad (\gamma = 0.023 \text{ MPa/m})$$

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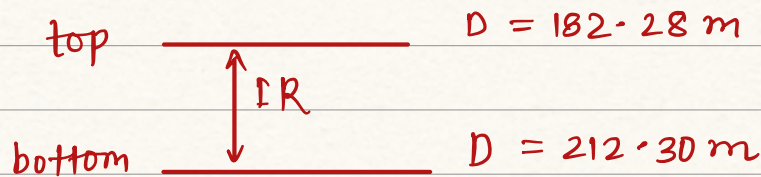
$B \rightarrow$ coefficient of thermal expansion

$$\text{sandstone} = 8 \times 10^{-6} / ^\circ\text{C}$$

$$\text{coal} = 30 \times 10^{-6} / ^\circ\text{C}$$

$E \rightarrow$ young's modulus (3.1 GPa)

For IR ;



$$\begin{aligned}\sigma_{H \text{ top}} &= \left(\frac{0.25}{0.75} \times 0.023 \times 182.28 \right) + \left(\frac{8 \times 10^{-6} \times 3.1 \times 10^2 \times 0.03}{0.75} \right) \times (1000 + 182.28) \\ &= 2.57 \text{ MPa}\end{aligned}$$

$$\sigma_{H \text{ bottom}} = 2.83 \text{ MPa}$$

$$\sigma_{H \text{ mean}} = \frac{2.57 + 2.83}{2} = 2.7 \text{ MPa}$$

$$\sigma_{mIR} = \left(\frac{2.7 + 0.388}{100} \right) \times 80 = 2.47 \text{ MPa}$$

For MR ;

$$\text{top} \quad \text{---} \quad D = 166.34 \text{ m}$$

$$\text{bottom} \quad \text{---} \quad D = 182.28 \text{ m}$$

$$\sigma_{H \text{ top}} = 2.72 \text{ MPa}$$

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$$\sigma_H \text{ bottom} = 2.86 \text{ MPa}$$

$$\sigma_H \text{ mean} = \frac{2.72 + 2.86}{2} = 2.79 \text{ MPa}$$

$$\sigma_{mMR} = \left(\frac{2.79 + 0.352}{100} \right) \times 66 = 2.073 \text{ MPa}$$

$$W_s = 324 \times (150)^{-0.43} \times (212.48)^{-1.38} \times \sqrt{\frac{2.47 \times 30.02}{2045 \times 10 \times 10^{-6}}} \times \sqrt{\frac{2.073 \times 15.94}{2050 \times 10 \times 10^{-6}}}$$

$$= 55.77 \text{ m}$$

$$S_v = 0.0014 D^{0.54} h^{1.96} (FL)^{1.36} \underbrace{\sigma_{cIR}^{0.76} \sigma_{cMR}^{0.04} \sigma_{cCoal}^{-0.34}}_{\text{rockmass}} T_R^{0.17} \left[\frac{E_{IR} \times E_{MR}}{E_c^2} \right]^{-0.47}$$

$D \rightarrow$ depth

$h \rightarrow$ extraction height

$FL \rightarrow$ face length

$$\sigma_{cIR} = 4.008 \text{ MPa}$$

$$\sigma_{cMR} = 4.118 \text{ MPa}$$

$$\sigma_{cCoal} = 4.632 \text{ MPa}$$

$$T_R = \text{thickness ratio} = \frac{T_{MR}}{T_{IR}} = \frac{15.94}{30.02}$$

$$E = \text{modulus of elasticity} = 0.31 \times \sigma_c (\text{intact})$$

$$S_v = 0.0014 \times (212.48)^{0.54} \times (3)^{1.96} \times (150)^{1.36} \times (4.008)^{0.76} \times (4.118)^{0.04} \times (4.632)^{-0.34} \times \left(\frac{15.94}{30.02} \right)^{0.17} \times \left[\frac{0.31 \times 10.02 \times 0.31 \times 12.46}{(0.31 \times 23.16)^2} \right]^{-0.47}$$

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Finally !!!

$$DRFP = 25.583 + 0.042 S_v + 0.146 W_s$$

↑ ↑
Put values here